NAME ______________________ DATE ____________ PERIOD ________

Paper Practice 6.4

Solving Absolute Value Inequalities

WORKED KEY – Show all of your work in the space provided. Graph each solution set on a number line. Remember: < gives AND, > gives OR.

1. Solve each equation. (Warm-Up Review)

a. $|x| = 4$ ANSWER: $x = 4$ or $x = -4$

b. $|x - 3| = 7$ ANSWER: $x = 10$ or $x = -4$

c. $|2x + 8| = 10$ ANSWER: $x = 1$ or $x = -9$

2. Solve and graph the solution set. (Less Than)

a. $|m + 5| < 3$ ANSWER: $-3 < m + 5 < 3$, so $-8 < m < -2$

b. $|m - 7| < 7$ ANSWER: $0 < m < 14$

c. $|x + 4| < 6$ ANSWER: $-10 < x < 2$

3. Solve and graph the solution set. (Greater Than)

a. $|2m - 9| > 13$ ANSWER: $2m - 9 > 13 \rightarrow m > 11$; $2m - 9 < -13 \rightarrow m < -2$. So $m < -2$ or $m > 11$.

b. $|3m + 12| \geq 15$ ANSWER: $3m + 12 \geq 15 \rightarrow m \geq 1$; $3m + 12 \leq -15 \rightarrow m \leq -9$. So $m \leq -9$ or $m \geq 1$.

4. Solve. Watch for the special cases. (Special Cases)

a. $|n - 1| < -5$ ANSWER: No solution – an absolute value is never negative.

b. $|n + 11| < -13$ ANSWER: No solution.

c. $|n - 6| \geq -5$ ANSWER: All real numbers – an absolute value is always ≥ 0, so it is always ≥ -5.

NAME ______________________ DATE ____________ PERIOD ________

5. Solve. (Mixed)

a. $|x - 2| \le 5$ ANSWER: $-3 \le x \le 7$

b. $|4x| > 12$ ANSWER: $x < -3$ or $x > 3$

6. Isolate the absolute value first, then solve. (Isolate First)

a. $|2x + 1| - 3 < 8$ ANSWER: $|2x + 1| < 11 \rightarrow -11 < 2x + 1 < 11 \rightarrow -12 < 2x < 10 \rightarrow -6 < x < 5$

b. $3|x - 4| \ge 18$ ANSWER: $|x - 4| \ge 6 \rightarrow x \ge 10$ or $x \le -2$

7. Error Analysis – Gus solved $|x + 3| > 5$ and wrote $-8 < x < 2$. Identify his mistake and give the correct solution. (Error Analysis)
ANSWER: He used the AND rule, which belongs to <. A > inequality gives OR: $x < -8$ or $x > 2$.

8. Write an absolute value inequality whose solution is $-3 \le x \le 9$. (Write One) ANSWER: Centre 3, distance 6: $|x - 3| \le 6$

9. Write an absolute value inequality whose solution is $x < -1$ or $x > 7$. (Write One)
ANSWER: Centre 3, distance 4: $|x - 3| > 4$

10. A machine fills bottles to 500 mL with a tolerance of 8 mL. Write and solve an absolute value inequality for the acceptable volume v. (Apply) ANSWER: $|v - 500| \le 8$, so $492 \le v \le 508$ mL.

11. Explain why $|x| < -2$ has no solution but $|x| > -2$ is true for every real number. (Explain)
ANSWER: An absolute value is a distance, so it is never negative. It can never be less than −2, and it is always greater than −2.